AWS SAA-C03 Exam Notes: RDS, Aurora, and ElastiCache

1. Amazon RDS Overview

- **What is it?** Managed Relational Database Service. AWS manages the OS, patching, backups, and hardware. You manage the app optimization and schema.
- **Supported Engines:** Postgres, MySQL, MariaDB, Oracle, Microsoft SQL Server, IBM Db2.
- **Instance Types:**
  - *Standard* (m-class): General purpose.
  - *Memory Optimized* (r-class, x-class): Great for production DBs with heavy loads.
  - *Burstable* (t-class): Dev/Test or low workloads.
- **Storage Types:**
  - **General Purpose SSD (gp2/gp3):** Cost-effective. Transactional workloads.
  - **Provisioned IOPS SSD (io1/io2/Block Express):** High performance, low latency, mission-critical. **Supports Multi-Attach** (io2 only on EBS, not RDS directly in the same way, but RDS uses io1/io2 for intense I/O).
  - **Magnetic:** Legacy (avoid in exam answers unless "cost" is the *only* factor and performance doesn't matter).
- **Auto Scaling:** RDS Storage Auto Scaling can increase storage automatically without downtime (you must set a maximum limit).

2. RDS Read Replicas vs Multi-AZ (High Availability)

This is a guaranteed exam topic. Know the difference cold.

Feature	Read Replicas	Multi-AZ (Standby)
Primary Goal	**Scalability** (Offload reads)	**High Availability / Disaster Recovery**
Replication	**Asynchronous** (Eventual consistency)	**Synchronous** (Strong consistency)
Active/Passive	All active (Primary writes, Replicas read)	Active Primary / Passive Standby
Failover	Manual (must promote replica to primary)	**Automatic** (DNS points to standby)
Traffic	Serves **READ ONLY** traffic	**NO TRAFFIC** served by standby (it is idle)
Scope	Can be **Cross-AZ** or **Cross-Region**	**Within Region** (spans 2 AZs)

Feature	Read Replicas	Multi-AZ (Standby)
Backups	Backups typically taken from Primary	Backups taken from Standby (avoids I/O suspension)

> **Important Numbers:**
>
> - **RDS Read Replicas:** Max **15** replicas for ASG/MySQL/MariaDB/PostgreSQL. Max **5** for Oracle.
> - **Multi-AZ:** Typically **1 Standby** instance (2 copies total).
> - **Note on Multi-AZ Cluster:** There is a newer deployment option called "Multi-AZ with two readable standbys" for MySQL/PostgreSQL. However, in standard exam questions, if they just say "Multi-AZ," assume the classic **Active/Passive** setup where the standby cannot accept traffic.

- **Exam Tip:** If a question asks to "improve performance of a read-heavy analytics application," choose **Read Replicas**.
- **Exam Tip:** If a question asks for "fault tolerance" or "durability" in case of a data center failure, choose **Multi-AZ**.

3. RDS Custom (Oracle & Microsoft SQL Server)

- **Use Case:** You need to migrate a legacy application that requires **access to the underlying Operating System (OS)** or requires you to install custom software/agents on the DB server.
- **Management Model:**
  - Standard RDS: You **cannot** SSH/RDP into the instance.
  - **RDS Custom:** You **CAN** SSH/RDP into the instance.
- **Responsibility Shift:** Because you have OS access, **you** are responsible for OS patching. AWS will pause automation if you make changes that interfere with their hooks.
- **Engines:** Only available for Oracle and Microsoft SQL Server.

4. Amazon Aurora

- **Architecture:** AWS-native relational DB (MySQL & PostgreSQL compatible).
- **Storage (The "Magic" Part):**
  - Data is stored in a **shared Cluster Volume** (virtual storage), not local EBS.
  - **3 AZs, 6 Copies:** Aurora writes 6 copies of your data across 3 AZs.
  - **Self-Healing:** Automatically isolates bad disk blocks.
  - **Auto-Expanding:** Storage grows automatically (up to 128 TB).
- **Availability & Replicas:**
  - Write availability: Needs 4 of 6 copies.
  - Read availability: Needs 3 of 6 copies.

    - **Aurora Replicas:** Max **15** replicas (Auto Scaling enabled).
    - If Master fails, an Aurora Replica is promoted automatically (very fast, < 30s)
- **Endpoints (Crucial for Exam):**
    - **Cluster Endpoint (Writer Endpoint):** Points to the ONE primary instance. Use for writes.
    - **Reader Endpoint:** Load balances traffic across ALL read replicas. Use for reporting/analytics.
    - **Custom Endpoint:** A subset of instances (e.g., "Analytics Nodes" vs "Web Nodes").

5. Amazon Aurora - Advanced Concepts

- **Aurora Serverless (v1 & v2):**
    - Automated scaling for **intermittent or unpredictable workloads**.
    - No instance provisioning. Pay per second (v2) or minute (v1).
    - *Exam Use Case:* "A startup has a database that is used heavily for 2 hours a day but is idle the rest of the time." -> **Aurora Serverless**.
- **Aurora Global Database:**
    - **Physical Replication:** Replicates at the storage layer (latency < 1 second).
    - **Secondary Regions:** Supports up to **10 secondary regions** (check exam version, formerly 5).
    - **Cross-Region DR:** Promote a secondary region to primary in < 1 minute.
    - *Exam Use Case:* "Mission-critical app requires RPO < 1 second and RTO < 1 minute across regions."
- **Aurora Backtrack (MySQL Only):**
    - Rewind the database to a specific point in time without restoring from a backup.
    - Exclusive to Aurora MySQL.
- **Aurora Machine Learning:**
    - Enables you to add ML-based predictions to your applications via SQL.
    - **Integration:** Direct integration with **Amazon SageMaker** (for any ML model) and **Amazon Comprehend** (for sentiment analysis).
    - *Exam Benefit:* No need to build complex ML pipelines; just call a SQL function.
- **Babelfish for Aurora PostgreSQL:**
    - **Goal:** Run Microsoft SQL Server applications on Aurora PostgreSQL with minimal code changes.
    - **How it works:** It allows Aurora PostgreSQL to understand T-SQL (Transact-SQL) and listen on port 1433.
    - *Exam Use Case:* "Migrating legacy SQL Server app to open-source AWS DB with minimal refactoring."

6. RDS & Aurora - Backups, Monitoring & Maintenance

- **Backup & Restore Architecture (Exam Critical):**

    - **RDS Backup:**
        - **Snapshots:** Traditional snapshots of the storage volume.
        - **Performance:** In Single-AZ, I/O may suspend briefly. In Multi-AZ, backup is taken from the Standby (no impact).
        - **Retention:** 1-35 days for automated. Manual snapshots kept indefinitely.
    - **Aurora Backup:**
        - **Continuous Backup:** Backups are essentially continuous and streaming to S3.
        - **Performance: Zero performance impact** on the database (no I/O freeze).
        - **Restore:** Point-in-Time Recovery (PITR) is faster than RDS.
- **Database Cloning (Aurora Only):**
    - **Concept:** Creates a new cluster from an existing one using **Copy-on-Write** protocol.
    - **Speed:** Instant (does not copy full data initially).
    - **Cost:** You only pay for the *changes* made to the new clone.
    - *Exam Use Case:* "Need to create a Staging environment using Production data for testing immediately without high storage costs."
- **Monitoring Tools (Know the difference!):**
    - **CloudWatch:** *Hypervisor* metrics. CPU, Network, Disk I/O.
        - *Limitation:* Can't see *what* query or process is consuming resources.
    - **Enhanced Monitoring:** *OS-level* agent.
        - *Key Metrics:* Memory, **Process List** (which process is using CPU?), Swap.
        - *Granularity:* Down to 1 second.
    - **Performance Insights:** *Database-level* dashboard.
        - *Key Metrics:* **DB Load (AAS)**, Wait Events, Top SQL Queries.
        - *Exam Scenario:* "Database is slow, CPU is 100%. Which SQL query is responsible?" -> **Performance Insights**.
- **Maintenance Windows:**
    - Required for engine version upgrades or OS patching.
    - **RDS Multi-AZ:** Standby patched first, then failover, then Primary patched. (Brief downtime for failover).

7. RDS Security

- **Encryption at Rest:** Enabled at creation time using **KMS** (AES-256). You cannot encrypt an existing unencrypted DB directly (Snap -> Copy as Encrypted -> Restore).
- **Encryption in Transit:** SSL/TLS is supported. Enforce using parameter groups (`rds.force_ssl=1` for Postgres/MySQL) or SQL commands.
- **IAM Database Authentication:**
    - Instead of a database password, use an IAM Token (valid for 15 mins).

    - Best for: Lambda functions or EC2 instances accessing RDS. Removes need for hardcoded credentials.
- **Security Groups:** Control **network** access (allow port 3306/5432 from App Security Group).

8. RDS Proxy

- **Problem:** Opening and closing DB connections is "expensive" (CPU/RAM). High-concurrency apps (like Lambda) can exhaust DB connections (aka "Connection Storms").
- **Solution:** RDS Proxy sits between the App and the DB.
    - **Connection Pooling:** Reuses existing connections.
    - **Failover Reduction:** Reduces Aurora failover time by up to 66% (bypasses DNS propagation delays).
- **Exam Keyword:** "Lambda," "Too many connections," "Connection timeout," "Reduce failover time."

9. ElastiCache Overview

- **What is it?** Managed In-Memory Caching (Redis & Memcached). Helps offload read-heavy workloads from the database.

Redis vs. Memcached (The Comparison)

Feature	Redis	Memcached
Data Types	**Complex** (Strings, Lists, Sets, Sorted Sets, Hashes, Bitmaps)	**Simple** (Strings, Objects)
Persistence	**Yes** (AOF/RDB) - Data survives reboot	**No** - Data lost on reboot
Availability	**Multi-AZ** with Auto-Failover	No Multi-AZ (Sharding only)
Scaling	Scale Read Replicas & Clustering	Multithreaded (Scale Up/Out)
Features	Pub/Sub, Geospatial, Transactions, **Leaderboards**	Simple Caching
Compliance	HIPPA, FedRAMP	HIPPA, FedRAMP

- **Exam Shortcut:**
    - Need "Leaderboards", "Sorting", "Persistence", or "Advanced Types"? **Choose Redis.**
    - Need "Simple multithreaded cache" or "Object caching"? **Choose Memcached.**

10. ElastiCache for Solution Architects (Strategies)

Caching Strategies

1. **Lazy Loading (Cache-Aside):**
   - App checks Cache. If miss, queries DB, writes to Cache, returns to user.
   - *Pros:* Only requested data is cached. Node failure not fatal.
   - *Cons:* "Cache Miss" penalty (3 network calls). Data can become stale.
2. **Write-Through:**
   - App writes to DB and Cache simultaneously.
   - *Pros:* Data never stale. fast reads.
   - *Cons:* Write penalty (slower writes). Wasted memory on data that is never read.

Use Case: Session Store

- **Scenario:** You have a stateless web application scaling across multiple EC2 instances. Where do you store the user's login session?
- **Option A:** DynamoDB (Great, but higher latency than cache).
- **Option B: ElastiCache (Redis)**.
  - *Why?* Sub-millisecond latency. You can set a TTL (Time to Live) to auto-expire sessions.

Caching at Scale

- If you see "Global" or "Cross Region" caching requirements -> **Redis Global Datastore**.

11. Crucial Exam Extras (Must Know!)

AWS Secrets Manager Integration

- **Problem:** Hardcoding DB passwords in code is a security violation.
- **Solution:** Store credentials in **AWS Secrets Manager**.
- **Key Feature: Automatic Rotation.** Secrets Manager can automatically rotate the password in RDS and give the new one to the application without you doing anything.
- *Exam Keyword:* "Rotate credentials," "Remove hardcoded passwords," "Retrieve secrets automatically."

RDS Backups vs. Snapshots

- **Automated Backups:**
  - Enabled by default.
  - Retention: 1 to 35 days (Set to 0 to disable).
  - Point-in-Time Recovery (PITR) capable.
  - *Deleted when you delete the instance* (unless you check "Retain final backup").
- **Manual Snapshots:**
  - User initiated.

    - Retention: **Indefinite** (Never expire).
    - *Persist even if the original RDS instance is deleted.*
- *Exam Scenario:* "Company needs to keep database backups for 5 years for auditing." -> Use **Manual Snapshots** (or AWS Backup service), as Automated only goes to 35 days.

Parameter Groups: Static vs. Dynamic

- **Dynamic Parameters:** Changes apply **immediately** (no reboot).
- **Static Parameters:** Changes require a **MANUAL REBOOT** of the database to take effect.
- *Exam Scenario:* "You updated a parameter to enable logging, but it's not working." -> Check if it was a static parameter and if the instance was rebooted.

Database Migration Service (DMS)

- **What is it?** Migrates databases to AWS while the source DB remains operational (minimal downtime).
- **Schema Conversion Tool (SCT):** Used *before* DMS if engines are different (e.g., Oracle -> Aurora PostgreSQL) to convert schema/procedures.
- **Homogeneous:** Same engine (MySQL -> RDS MySQL).
- **Heterogeneous:** Different engine (SQL Server -> Aurora). Use SCT + DMS.